

EDUCATION

Instituto Tecnológico

San Carlos, Costa Rica

Bachelor of Science (B.S.) in Computer Engineer, GPA: 3.88/4.0

Feb 2023 – Present

- **Concentrations:** Artificial Intelligence & Web Development, Programming Languages (Python, Rust, Java, C++), Algorithm Design, Database Management (MySQL) **Soft Skills (Communication, Teamwork)**
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Objects & Software Design, Computer Organization, Design of Algorithms, Database Systems, Statistics, Applied Combinatorics, Programming Languages, Probability, Project Management, Compilers & Interpreters, Software Quality Assurance
- **Languages:** Spanish (Native), English (Upper-Intermediate / CEFR B2 – self-assessed)

WORK EXPERIENCE

Research Assistant in Artificial Intelligence

San Carlos, CR

Student Software Engineer

June 2023 – May 2025

- Developed multiple software solutions leveraging advanced face anti-spoofing algorithms to accurately detect and block counterfeit and AI-generated facial images, successfully identifying fake OLED screen replays and other complex spoofing scenarios that typically defeat standard models.
- Developed a software application leveraging model embeddings to recognize each registered employee, eliminating the need for manual clock-ins and thereby reducing long queues, saving time, and improving organizational efficiency.
- Created a software solution for verifying individual authenticity by analyzing eye blinks through eye detection, resulting in a 35% reduction in false image recognition.
- Researched and implemented an AI-driven, image-based person recognition and contact management system designed to identify conversation partners and automatically provide concise summaries of past interactions—preventing forgetfulness and enabling smoother, more informed conversations.

Statistical Book Created with R

San Carlos, CR

Mathematics Instructor's Personal Project

August 2023 – January 2024

- Developed proficiency in diverse ggplot2 visualization techniques in R, ensuring deeper understanding and more insightful data analysis.
- Authored and compiled a comprehensive statistical book utilizing R for advanced data analysis and visualization.
- Employed LaTeX for precise mathematical typesetting, ensuring clear and accurate presentation of formulas and equations.
- Utilized Markdown to structure and format textual content and sections, enhancing readability and organization.
- Incorporated various libraries to generate informative graphs and charts, effectively illustrating diverse data insights.

PROJECTS

- **PuraCompra: E-commerce Intermediary Platform:** Developed an intermediary platform to streamline online purchases across Amazon, Best Buy, Micro Center, Walmart, and Gymshark; built a Node.js and Express backend with MySQL for robust database operations; created a React.js and Bootstrap frontend featuring shopping cart management, order status tracking, and user role differentiation; implemented promotion, review, and rating functionalities to enhance the shopping experience; and delivered a clean, minimalistic, accessible, and responsive design that overcomes cross-border import hurdles while providing intuitive package management and tracking. Available at: [GitHub](#)
- **Ticket Booking Server in Rust:** Developed a high-performance ticket booking server using Rust, leveraging its efficiency and stability. Designed a socket-based communication system to handle real-time ticket transactions and seat availability updates. Implemented an algorithm to determine optimal seating based on visibility percentage and proximity to desired locations. Ensured robust error handling and concurrency management to maintain server reliability under high load. Available at: [GitHub](#)
- **iFixIt:** Developed iFixIt, a web and mobile platform for automotive workshop management enabling administrators to register vehicles by license plate, log repair details (photos, audio notes, and descriptions) organized by date, and set service pricing; and empowering customers via the mobile app to track order progress, view performed work, locate nearby workshops with Google Maps integration, and schedule vehicle pickup. Available at: [GitHub](#) **Live Demo:** [Static website deployment for project showcase](#)

- **Web Scraping App:** Tackled the tedious process of manually hunting for game promotions by creating an automated Scrapy crawler to extract pricing data from G2A, Instant Gaming, and PS4 Store; stored results in a JSON-based mini-database for instant lookups; built a Flask backend to serve up-to-date pricing APIs; and designed a Vue.js frontend for a responsive, user-friendly interface with automatic updates. Available on GitHub: [Web-Scraping-Game-Crawler](#).
- **Statistical Book Created with R:** Authored a comprehensive statistical book that integrates advanced data analysis and visualization techniques using R. Combined LaTeX for mathematical notation, Markdown for structured content, and various R libraries to generate high-quality statistical graphs. The book serves as a resource for students and researchers, offering in-depth insights into statistical methodologies. Available at: [Statistical Book](#).
- **Research Assistant in Artificial Intelligence:** Developed a software solution for detecting real versus fake individuals using Face Anti-Spoofing techniques. Trained YOLO models to recognize various types of spoofed images related to facial authentication and object detection. Implemented eye detection algorithms to verify authenticity by analyzing eye blinks. Additionally, conducted research on Vision Transformer architectures for enhanced image recognition capabilities. More details at: [AI Research Project](#), [AI Face Detection](#)

EXTRACURRICULAR ACTIVITIES

Robotics: Engaged in an advanced robotics course focused on designing, constructing, and programming robotic systems. Acquired hands-on experience in both mechanical assembly and software development to enhance the functionality and performance of robotic systems.

Computer Building and Repair: Extensive experience in assembling and troubleshooting computers. Successfully built numerous custom PCs and addressed various technical issues through comprehensive repairs and upgrades.

SKILLS

Programming Languages: Python, Java, JavaScript, C++, C, Rust, LaTeX, R, Go

Version Control & Collaboration: Git

Web Development: Node.js, React, HTML/CSS, Tailwind

Databases: MySQL, MongoDB, PostgreSQL

Backend Frameworks: Flask, Express.js

Cloud Platforms: AWS, Azure, Google Cloud

Containerization & Virtualization: Docker

Operating Systems: Linux, macOS, Windows

Data Science & Machine Learning: PyTorch, TensorFlow, OpenCV

Documentation: Pandoc, LaTeX, Markdown, Typst

Productivity Tools: Notion

REFERENCES

Ph.D. Abel Méndez Porras: *Data Processing, Image Processing & Artificial Intelligence*

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